

CNN-Based Plant Leaf Disease Classification System

Short Description of Methods

1. Transfer Learning

Transfer learning is a deep learning technique where a pre-trained CNN model such as **ResNet50** or **MobileNetV2** is reused for a new classification task. These models are already trained on large datasets like ImageNet and can extract important image features such as edges, textures, and shapes. In this project, the final classification layers are modified to classify healthy and diseased plant leaves.

2. Data Augmentation

Data augmentation is used to increase dataset diversity and reduce overfitting. It generates modified versions of training images using techniques such as:

- Rotation
- Flipping
- Zooming
- Shearing
- Brightness adjustment

This improves the model's generalization ability and robustness.

3. Optimizer Comparison

Optimizers are algorithms used to update model weights during training. Two optimizers are compared:

- **Adam Optimizer:** Fast convergence and adaptive learning rate.
- **SGD Optimizer:** Uses gradient descent with momentum for stable learning.

The comparison helps determine which optimizer provides better accuracy and lower loss.

4. Accuracy and Loss Curves

Training and validation accuracy/loss curves are plotted to monitor model performance across epochs. These graphs help identify:

- Overfitting
- Underfitting
- Learning behavior of the model

5. Confusion Matrix

A confusion matrix is generated to evaluate classification performance. It shows:

- Correctly classified samples
- Misclassified samples
- Class-wise prediction performance

6. Misclassified Class Identification

The system identifies the classes that are most frequently misclassified. This helps analyze difficult disease categories and improve dataset quality or model performance.

Algorithm Steps

Algorithm: Plant Leaf Disease Classification using CNN

Step 1: Dataset Collection

1. Download the PlantVillage dataset.
2. Organize images into class folders containing healthy and diseased leaf images.

Step 2: Data Preprocessing

1. Resize all images to a fixed size (e.g., 224×224).
2. Normalize pixel values.
3. Split dataset into:
 4. Training set
 5. Validation set
 6. Testing set

Step 3: Apply Data Augmentation

1. Apply image transformations:
 2. Rotation
 3. Horizontal flip
 4. Zoom
 5. Shear
6. Generate augmented images during training.

Step 4: Load Pre-trained CNN Model

1. Import ResNet50 or MobileNetV2 model.
2. Load ImageNet pre-trained weights.
3. Remove the final classification layer.
4. Add:
 5. Global Average Pooling layer
 6. Dense layer
 7. Softmax output layer

Step 5: Compile the Model

1. Select loss function:
 2. Categorical Crossentropy
3. Select optimizers:
 4. Adam
 5. SGD
6. Define evaluation metric:
 7. Accuracy

Step 6: Train the Model

1. Train model using augmented training data.
2. Validate performance on validation dataset.
3. Save best model weights.

Step 7: Compare Optimizers

1. Train the model separately using Adam and SGD.
2. Compare:
 3. Accuracy
 4. Loss
 5. Training time

Step 8: Plot Performance Curves

1. Plot:
2. Training Accuracy vs Validation Accuracy
3. Training Loss vs Validation Loss
4. Analyze convergence behavior.

Step 9: Evaluate Model

1. Test the model on unseen test images.
2. Generate confusion matrix.
3. Calculate:
4. Accuracy
5. Precision
6. Recall
7. F1-score

Step 10: Identify Misclassified Classes

1. Find incorrectly predicted images.
2. Count misclassifications per class.
3. Display the most confusing disease categories.

Step 11: Final Prediction

1. Input a new plant leaf image.
2. Predict whether the leaf is healthy or diseased.
3. Display predicted class label with confidence score.

Expected Output

- Accurate classification of plant leaf diseases
- Comparison of ResNet50 and MobileNetV2 performance
- Optimizer performance comparison
- Accuracy/loss graphs
- Confusion matrix visualization
- Identification of most misclassified disease classes